

TCP - The Chess Project

A traditional chess game that can be played online, either P2P or on a centralized server.

Link to Youtube demonstration

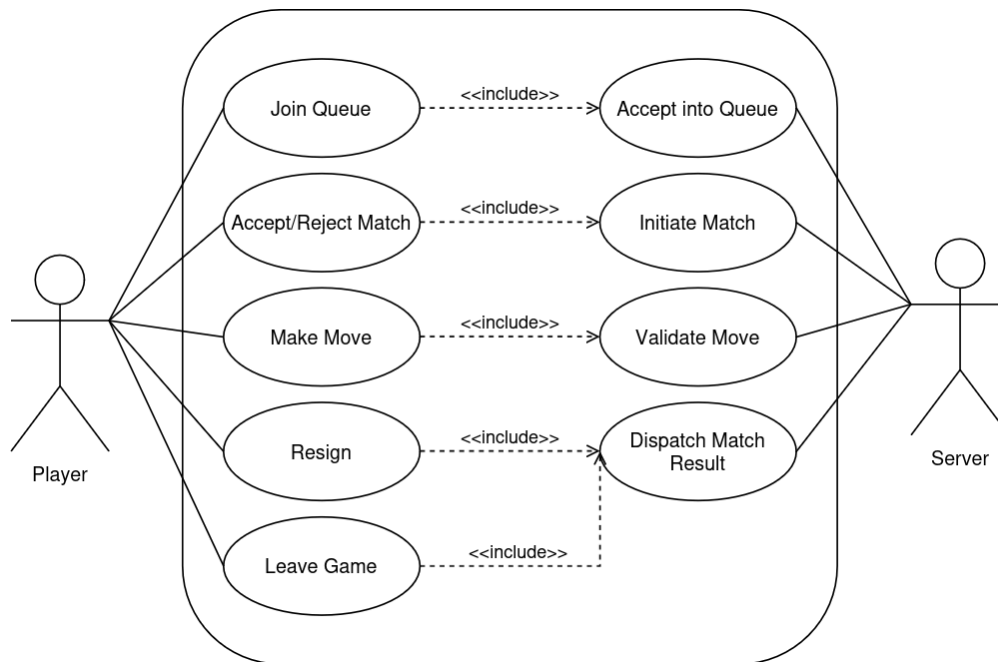
<https://www.youtube.com/watch?v=TV-5PNAcLSI>

Name of team: 3 Idiotai

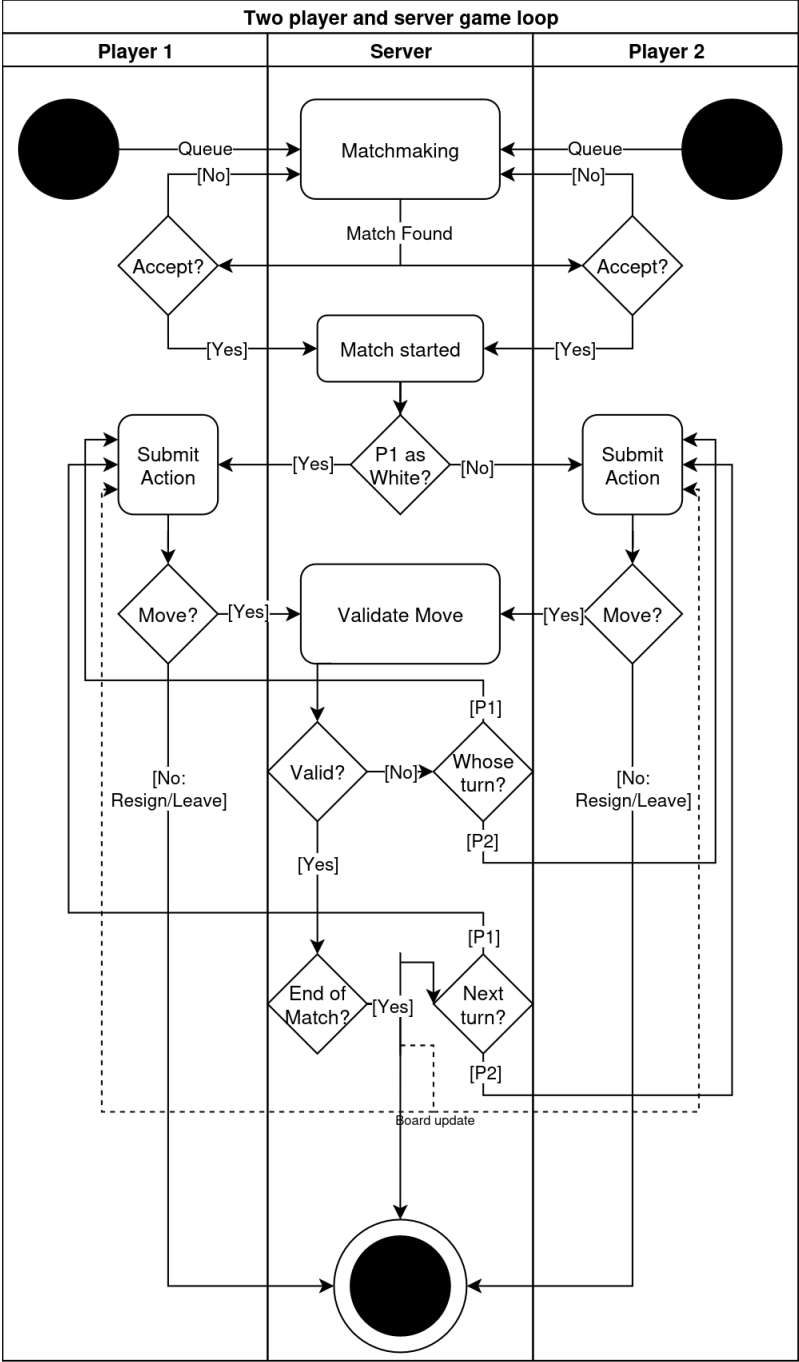
Member list (Role):

- Aringas Civilka (Facilitator, Architect, Builder, Exhibitor)
- Adomas Pimpė (Architect, Builder, Critic)
- Jonas Kirkilovskis (Architect, Builder, Critic)
- Girius Frankonis (Designer, Builder, Architect)

Use case diagram

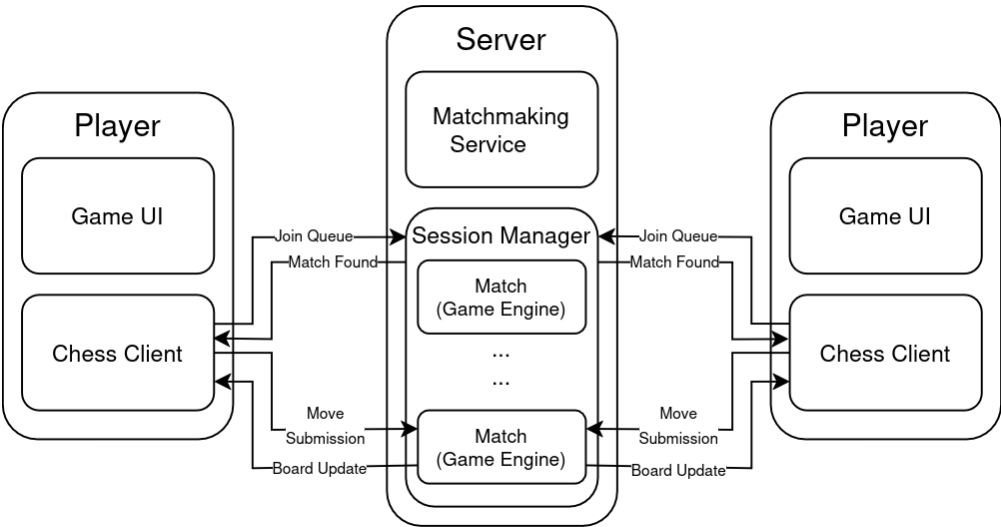


Activity diagram

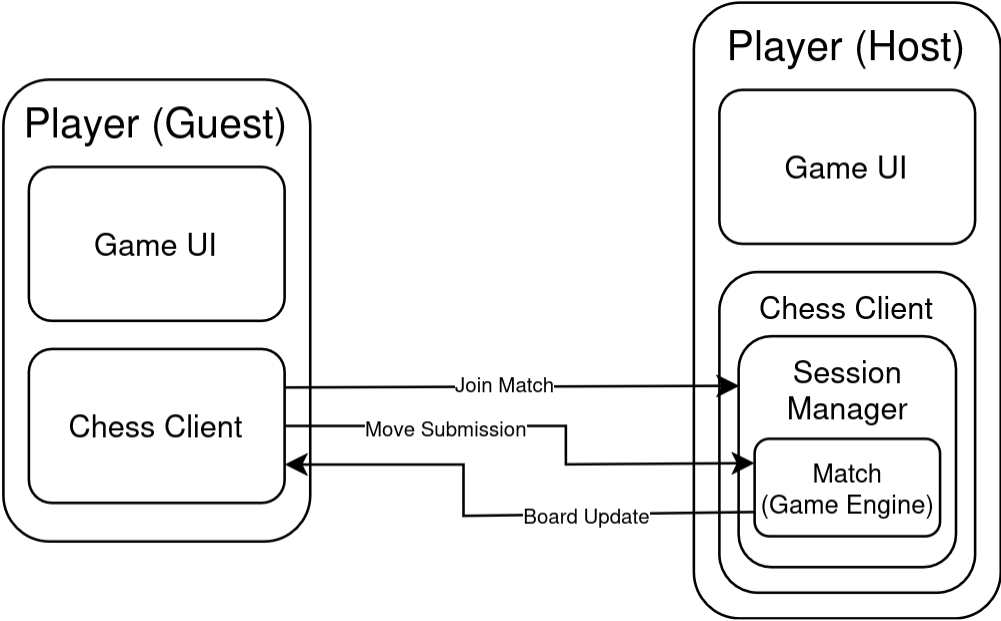


High level architecture of the system

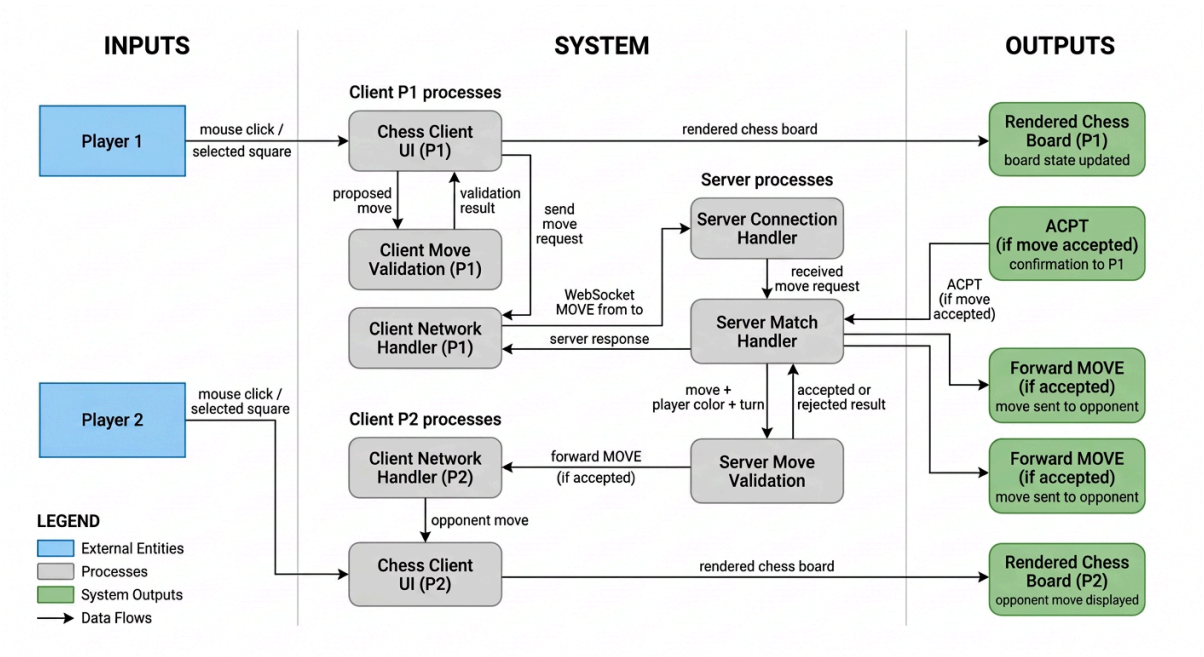
Centralized Server



Peer to Peer



Data flow diagram

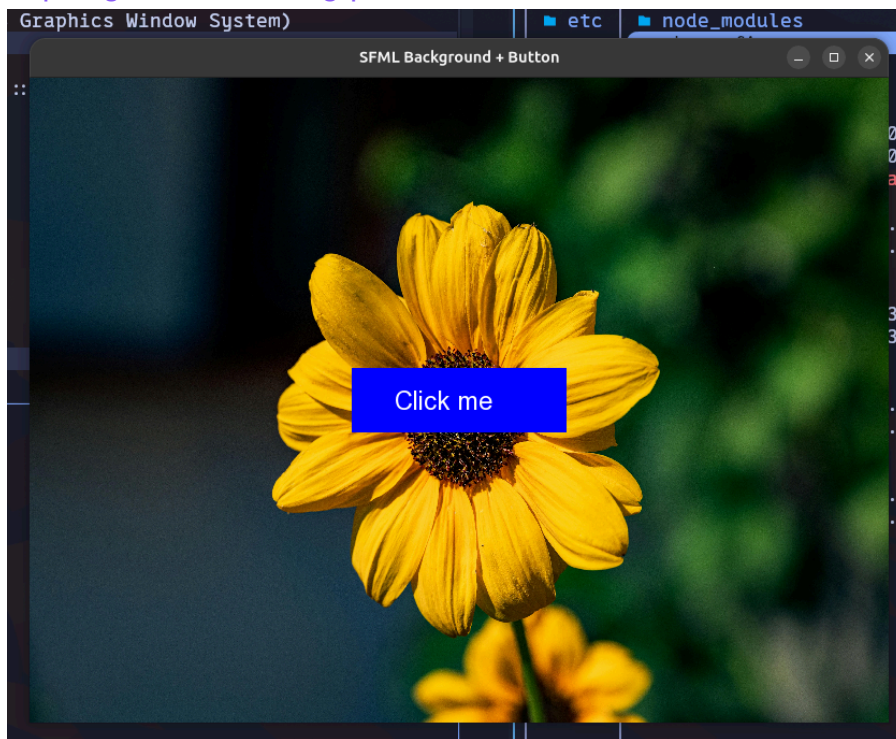


Technology stack:

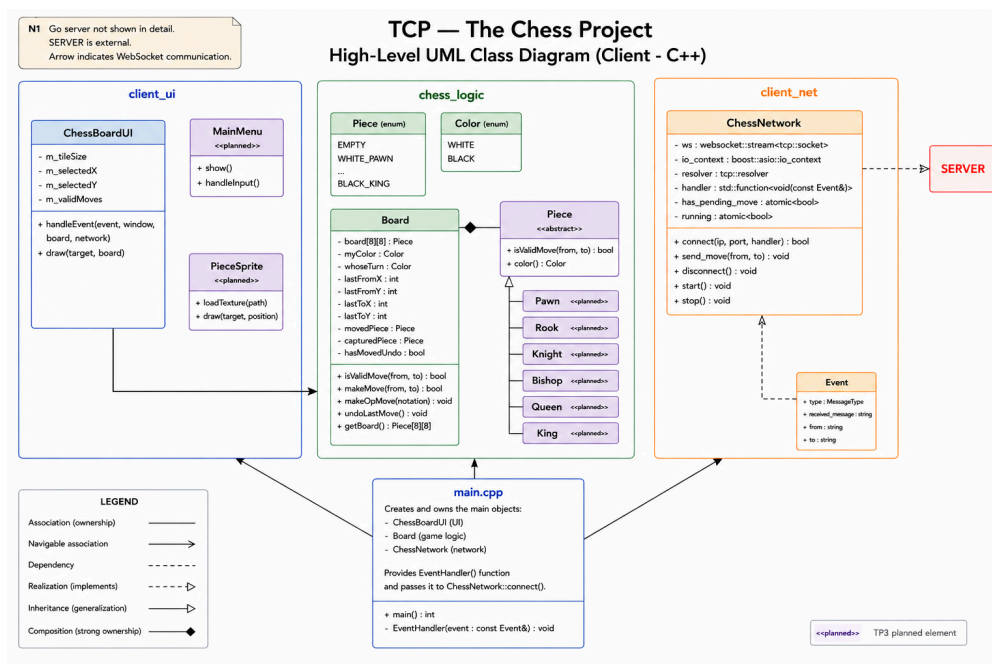
- Client side - C++ application using SFML for UI and ASIO/Beast for networking
- Server side - Golang using gorilla/websocket
- Version control - git + Github

Github repo & Proof of Concept:

<https://github.com/aringg10/TCP>



UML class diagram after TP2 (with plans for TP3)



Features implemented in TP3

- Server implements all classical pieces, checkmate, stalemate and resignation.
- UI with a menu, resign button and piece sprites.
- Client chess logic implements piece movement, castling and en passant.
- Client networking module talks to the server.
- Server-side time tracking.

Future work

- Client-side pawn promotion.
- King check recognition on the client.
- Client-side time tracking.
- Chat.
- Draw conditions such as the 50-move rule and insufficient material.

Project costs

TP1

	Time spent by						
Name	Person	Facilitator	Architect	Builder	Critic	Designer	Exhibitor
Aringas Civilka	4h40min	5min	4h	35min	0h	0h	0h
Adomas Pimpė	4h30min	0h	4h	0h	30min	0h	0h
Jonas Kirkilovskis	4h	0h	4h	0h	0h	0h	0h
Girius Frankonis	4h	0h	4h	0h	0h	0h	0h
Total:	17h10min	5min	16h	35min	30min	0h	0h

TP2

	Time spent by						
Name	Person	Facilitator	Architect	Builder	Critic	Designer	Exhibitor

	Time spent by						
Aringas Civilka	11h35min	10min	3h50min	5h35min	2h	0h	0h
Adomas Pimpė	10h	0h	4h	4h	2h	0h	0h
Jonas Kirkilovskis	10h15min	0h	3h15min	5h30min	1h30min	0h	0h
Girius Frankonis	8h	0h	30min	5h	1h	1h30min	0h
Total:	39h50min	10min	11h35min	20h5min	6h30min	1h30min	0h

TP3

	Time spent by						
Name	Person	Facilitator	Architect	Builder	Critic	Designer	Exhibitor
Aringas Civilka	17h20min	10min	30min	14h30min	2h	0h	10min
Adomas Pimpė	3h	0h	30min	2h	30min	0h	0h
Jonas Kirkilovskis	6h	0h	0h	5h30min	30min	0h	0h
Girius Frankonis	2h30min	0h	0h	2h	0h	30min	0h
Total:	28h50min	10min	1h	24h	3h	30min	10min

TP1 + TP2 + TP3

	Time spent by						
Name	Person	Facilitator	Architect	Builder	Critic	Designer	Exhibitor
Aringas Civilka	33h35min	25min	8h20min	20h40min	4h	0h	10min
Adomas Pimpė	17h30min	0h	8h30min	6h	3h	0h	0h
Jonas Kirkilovskis	20h15min	0h	7h15min	11h	2h	0h	0h
Girius Frankonis	14h30min	0h	4h30min	7h	1h	2h	0h
Total:	85h50min	25min	28h35min	44h40min	10h	2h	10min